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Multiple episodes of dike emplacement along the northwestern margin of the Superior Province, Manitoba

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Abstract. Three stable magnetization components A, B, and C have been isolated from a set of dikes that have been collectively referred to as the Molson swarm. Positive baked contact tests and square-shouldered thermal decay curves indicate that these components were acquired during three periods of dike emplacement. Dikes carrying A with a pole at 15.4°N and 96.5°W (A_{95}=4.1°, N=16 sites) have been dated by the U-Pb method at 1883 Ma and are closely related to the Paleoproterozoic Trans-Hudson Orogen. Ages of B with a pole at 27.1°N and 140.8°W (A_{95}=3.8°, N=30) and C with a pole at 40.7°N and 156.3°W (A_{95}=9.7°, N=19) are estimated at 2170-2120 Ma, on the basis of paleomagnetic characteristics that are distinct from the A component, the analysis of metamorphic history of the Pikwitonei Domain, and comparison with well-dated paleomagnetic poles from mafic igneous rocks in the Superior Province. The associated B and C dikes were emplaced in normal and reversed geomagnetic fields when the northwestern margin of the Superior Province experienced initial rifting prior to the Trans-Hudson Orogen. These earlier dikes may correlate with known 2094 Ma mafic magmatism documented in the lower Hurwitz Group of the Hearne Province. A stable magnetization component D with a pole at 20.9°N and 133.6°W (A_{95}=7.6°, N=30 samples) was also identified from unbaked granulites throughout the Pikwitonei Domain. This component predates all dike intrusion in the region, has an age range of 2200-2170 Ma, and may have been caused by earlier uplift of the northwestern Superior Province.

Introduction

The northwestern Archean Superior Province in Manitoba is divided into the Pikwitonei granulite domain and the lower-grade terrain of the Superior Province that consists of the Gods Lake and Molson Lake greenstone-granitoid domains. [*Weber*, 1990] (Figure 1). The granulites of the Pikwitonei Domain are believed to be a relatively deeply eroded section of the Gods Lake Domain and therefore represent uplifted crust [*Weber and Scoates*, 1978]. To the northwest of the Superior Province, the Paleoproterozoic Trans-Hudson Orogen dated at 1910-1760 Ma [*Baldwin et al.*, 1987; *Machado*, 1990] separates the Pikwitonei Domain from the Archean Hearne Province. The Churchill-Superior Boundary Zone (CSBZ) at the southeastern margin of the orogen includes the Thompson Belt, Split Lake Block, and Fox River Belt (Figure 1). The CSBZ represents a deformed continental margin dominated by Pikwitonei granulites retrogressed under amphibolite-grade conditions during development of the Trans-Hudson Orogen [*Weber and Scoates*, 1978; *Weber*, 1990].

A set of northeast trending dikes in the northwestern Superior Province (Figure 1) has long been referred to as the Molson dike swarm [*Fahrig et al.*, 1965; *Bell*, 1971; *Scoates and Macek*, 1978]. Although earlier paleomagnetic [*Ermanovics and Fahrig*, 1975] and radiometric studies [*Heaman et al.*, 1986a] suggested a single early Proterozoic intrusive episode, crosscutting relations between dikes [*Corkery*, 1985] indicate that the swarm may represent more than one age of intrusion.

In this paper we investigate the paleomagnetism of the Molson dikes as a function of distance from the CSBZ. An initial reconnaissance survey [*Halls et al.*, 1990] showed that westward within the Thompson Belt they became magnetically more unstable as the degree of metamorphism associated with the Trans-Hudson Orogen increased. Here we emphasize results from dikes within the Superior Province to the east and particularly within the Pikwitonei Domain where previous paleomagnetic experience, with granulites of the Kapuskasing zone [*Costanzo-Alvarez and Dunlop*, 1988; *Symons and Vandall*, 1990], suggests that host rock stability may allow definitive baked contact tests. Our principal objectives are to document differences in remanent magnetization within the Molson swarm that might define a multi-intrusive history and to interpret these data in terms of tectonic setting and their significance in the development of the Trans-Hudson Orogen.

Geology

Molson Dike Swarm

The swarm intrudes the lower-grade terrain of the Superior Province and the Pikwitonei Domain. Dikes found in the Thompson Belt and in the Split Lake Block are believed to be equivalent to the Molson dikes on the basis of chemical

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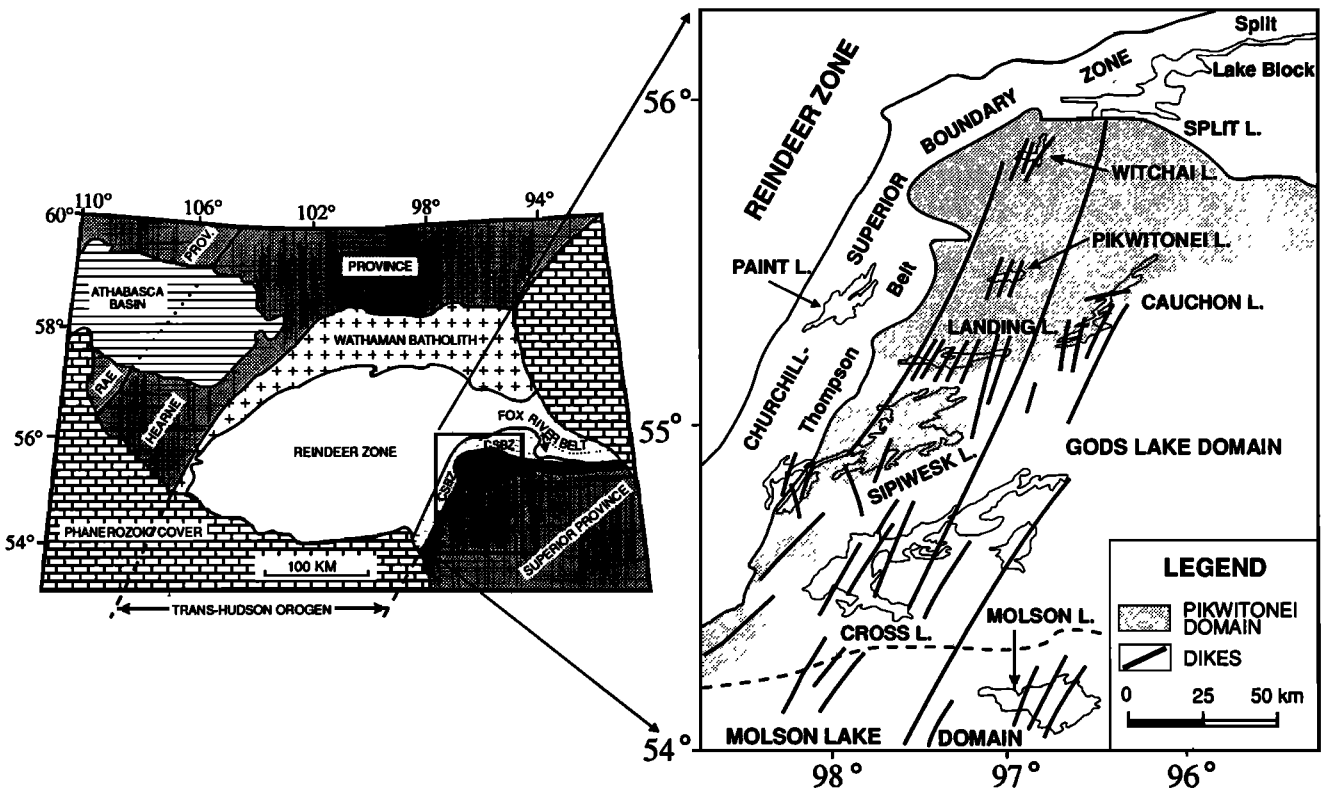


Figure 1. Simplified geological map of the northwestern part of the Superior Province and adjacent Trans-Hudson Orogen.

similarities [*Hubregtse*, 1977; *Scoates and Macek*, 1978; *Corkery*, 1985; *Weber*, 1990].

The majority of the dikes are subvertical with widths from less than 1 m to as much as 120 m. The dikes have a variable NE-NNW trend but are dominantly NNE-NNW close to the Thompson Belt. The frequency of dike intrusion increases toward and within the belt.

Molson dikes range from ultramafic to mafic in composition. Chemically, they display characteristics of tholeiitic basalts [*Ermanovics and Fahrig*, 1975] and komatiites [*Scoates and Macek*, 1978] which have rift-type chemical affinities [*Baragar and Scoates*, 1981; *Scoates*, 1981]. Olivine, orthopyroxene, clinopyroxene, plagioclase, and hornblende represent initial crystallization phases. Opaque minerals are mainly chromite, magnetite with ilmenite lamellae, and minor fine-grained sulphide minerals occurring as inclusions in hornblende, such as pyrrhotite, chalcopyrite, and pyrite [*Scoates and Macek*, 1978].

Within the Thompson Belt the Molson dikes were deformed and metamorphosed during the Trans-Hudson Orogen. The dikes southeast of the belt, however, generally escaped this event. In the lower-grade terrain of the Superior Province, deuteric alteration of olivine to magnetite and serpentine occurs and varying degrees of crystal fractionation features are developed [*Ermanovics and Fahrig*, 1975].

The earliest attempt at dating the Molson dikes was by the K-Ar whole rock method [*Wanless et al.*, 1970] which gave a range of ages from 1280 to 2393 Ma. *Ermanovics and Fahrig* [1975] attributed the scatter to various degrees of argon loss in the 1280-1700 Ma range, or excess in the case of the 2393 Ma age. *Cumming et al.* [1982] obtained a Pb isotope age of 2320 ± 30 Ma for primary sulfides (mainly pyrrhotite, pentlandite, and chalcopyrite) separated from four nickel sulphide deposits in the Thompson Belt. This age was interpreted as the time of emplacement of the host ultramafic rocks, and considered to be an estimate of the age of the Molson dike swarm [*Bell*, 1971; *Scoates and Macek*, 1978]. Later *Cumming and Scoates* [1985] reported a Pb-Pb whole rock age of 2450 Ma from a major dike (~45 m) at Landing Lake which has been interpreted as an age of emplacement of the dike. More recently, U-Pb zircon ages have been obtained by *Heaman et al.* [1986a] from the dikes in Cross Lake (1883 ± 2 Ma) and Cuthbert Lake (1882 ± 1.5 Ma), and from the Fox River sill (1883 ± 1.5 Ma) in the Fox River Belt (Figure 1). The 1883 Ma ages are generally regarded as the only time of the Molson dike intrusion. Despite the consistency of the 1883 Ma age, crosscutting relations in the Split Lake Block indicate that three ages of dike intrusion occurred between the Kenoran and Trans-Hudson Orogens [*Corkery*, 1985]. The tectonic setting and origin of the Molson swarm are debatable. It may have formed in (1) an initial rift associated with continental breakup prior to the development of the Trans-Hudson Orogen [*Green et al.*, 1985], (2) a foredeep environment resulting from buckling of Archean crust by Trans-Hudson thrust sheets advancing over the Superior Province [*Hoffman*, 1988], or (3) a back arc spreading setting prior to the collision of the Superior and Hearne Provinces [*Weber*, 1990].

Pikwitonei Granulite

The high-grade metamorphism of the granulites is dated between 2695 ± 2 Ma and 2637 ± 2 Ma (U-Pb zircon [*Heaman et al.*, 1986b]) with a metamorphic peak at ~2640

Ma [*Mezger et al.*, 1990]. The principal minerals of the granulites in the Pikwitonei Domain include clinopyroxene, plagioclase, hornblende, biotite, quartz, and magnetite. Garnet occurs locally in intermediate to mafic rocks. Olivine can also be seen locally in some ultramafic rocks. The main mineral assemblages of metamorphically retrogressed or reworked granulites in the CSBZ are characterized by garnet, quartz, biotite, amphibole, and feldspar. Incipient Paleoproterozoic metamorphism led to destruction of magnetite in the Archean granulites via biotite-, hornblende-, or garnet-forming reactions due to aqueous fluids penetrating the granulites [*Russell*, 1981; *Bleeker*, 1990].

Field and Laboratory Methods

Field Methods

Sampling areas were located at Cross and Molson Lakes in the lower-grade Superior Province, at Sipiwesk, Landing, Cauchon, Pikwitonei, and Witchai Lakes in the Pikwitonei Domain, and at Paint Lake in the Thompson Belt (Figure 1).

A total of 92 sites representing about 75 dikes were sampled. Samples were obtained at variable distances from dike margins wherever possible, with a majority taken from fine-grained parts or chilled margins. When available, host granulites adjacent to and remote from dikes were also collected for baked contact tests. In addition, fine-grained granulites, at least 1 km from any known dike, were sampled throughout Landing and Pikwitonei Lakes (sites CR1 and CR2, see Table 1) in an attempt to retrieve a pre-Molson dike remanence. Further sites were sampled from fine-grained and reworked granulites at Paint Lake to examine the influence of the Trans-Hudson Orogen on the paleomagnetism.

Samples were mostly field drilled at Molson, Landing, Pikwitonei, Paint, Cauchon, and Witchai Lakes and block sampled elsewhere. Sun and magnetic compasses were used for each sample orientation.

Table 1. Summary of Site Paleomagnetic Data

Site	S, deg	D, deg	W, m	RM	N_d/n_d	N_c/n_c	D, deg	I, deg	α_{95}	κ
					Cross Lake					
1	58	70N	0.7	ER	6/6	2/2	172.3	-42.6	35.2	3*
2	347	56W	1.0	ER	7/2	2/1	23.3	43.0	23.0	30*
3	73	53W	2.3†	UN	3/0	2/0				
4	120	67N	1.0	ER	5/1	--	102.2	76.9*		
5	57	83S	0.5†	B	7/4	--	243.5	68.4	6.7	191
6	36	58W	9.0†	ER	6/1	1/0	144.2	48.2*		
7	36	76W	2.3†	B	7/5	--	252.3	60.3	8.1	90
8	27	75W	1.1†	B	5/3	1/0	278.5	77.8	12.0	107
9	46	90	2.0	B	5/4	--	227.0	63.9	15.8	35
10	15	--	>3.0	B	4/1	--	270.9	49.8*		
11	30	--	>40.0	UN	6/0	--				
12	14	--	>50.0	ER	5/1	3/0	109.7	36.7*		
					Sipiwesk Lake					
13	23	88W	52.1	A	7/6	1/1	179.5	64.8	9.3	43
14	343	87W	>20.0	A	8/5	--	179.7	61.7	12.1	41
15	330	90	>23.0	A	9/7	--	198.1	71.7	9.0	46
16	336	--	27.0	A	3/0	3/1	188.7	73.4*		
17	330	90	>16.5	A	4/4	7/4	170.3	70.1	5.2	113
18		granulites		UN		5/0				
19	348	90	>20.0	A	9/9	2/1	179.5	67.2	4.4	121
20	25	85W	4.2	A	2/0	5/3	161.3	59.4	3.8	1063*
21	331	80W	>21.0	A	5/5	2/0	176.9	71.4	4.4	307
22	47	82E	>40.0	A	4/1	6/2	189.8	71.1	20.3	38*
23		granulites		UN		5/0				
24		granulites		UN		5/0				
					Molson Lake					
25	70	85E	59.1	B	12/5	3/0	241.6	60.8	18.6	18
26	35	85W	36.6	B	8/7	--	249.8	73.3	3.8	256
27	55	90	>36.6	B	13/7	--	236.1	80.5	3.3	338
					Landing Lake					
28	30	90	71.0	B	12/10	2/1	260.5	48.6	5.7	
29	46	--	45.0	B	4/3	--	262.5	59.4	11.1	125
30	40	90	15.0	B	7/4	5/4	253.1	73.7	6.1	82

Table 1. (continued)

Site	*S*, deg	*D*, deg	*W*, m	RM	N_d/n_d	N_c/n_c	*D*, deg	*I*, deg	α_{95}	κ
31		granulites		A		5/1	190.0	68.0*		
32	15	70W	>51.2	A	12/10	5/3	180.0	70.6	5.4	60
33	10	90	--	C	10/7	4/2	64.2	-57.7	10.3	26
34	32	--	>9.1	B	6/5	--	257.5	55.6	19.1	17
35	40	90	--	B	8/8	2/2	251.7	65.7	8.7	32
36	40	90	27.4	C	8/4	1/1	54.2	-58.7	13.7	32
37	40	90	67.0	A	7/5	6/3	167.8	70.1	6.9	65
38	6	82W	78.0	A	6/4	1/1	172.0	65.3	9.3	69
39	30	80W	10.4	A	5/5	--	190.4	64.5	8.2	88
40	26	80W	8.5	A	6/6	--	185.8	67.0	8.5	63
41	27	--	--	A	6/6	--	164.9	69.9	6.5	108
CR1		granulites		D		13/9	73.6	-60.9	16.3	13
					Witchai Lake					
42	45	90	3.1	C	7/7	3/1	112.3	-73.1.	5.7	96
43	339	90	4.7	C	6/5	7/2	40.6	-71.2	9.6	40
44	333	90	8.0	C	7/6	2/1	90.1	-65.7	1.7	1164
45	35	90	95.0	ER	5/0	2/1	104.8	62.2*		
46	69	90	>2.2	C	14/6	2/0	146.0	-69.8	16.7	17
47	92	87S	1.8	C	9/7	4/0	134.7	-59.7	9.4	43
48	60	78E	1.4	C	9/6	4/0	131.6	-63.3	8.4	65
49	63	56W	5.0	B	6/6	7/6	233.8	62.3	4.7	87
50	59	65E	15.0	B	7/7	3/2	240.9	62.6	5.7	82
51	53	90	10.6	A	6/6	2/2	174.1	70.7	7.6	54
52	6	65W	>20.0	B	6/6	5/5	227.3	63.4	4.9	87
					Pikwitonei Lake					
53	11	90	16.0	B	7/7	3/2	256.2	64.1	5.9	78
54	14	90	12.0	B	6/6	--	234.8	65.4	5.1	176
55	--	--	>60.0	C	9/6	2/0	56.3	-60.2	5.6	145
56	--	--	>25.0	B	8/8	3/0	254.1	58.8	6.8	68
57	345	90	22.0	B	5/3	5/4	256.2	69.2	7.5	65
58	--	--	--	B	5/4	--	235.2	62.3	21.2	20
59	65	79N	18.5	C	7/3	4/2	109.4	-50.1	15.2	26
60	35	90	38.0	B	7/5	6/5	247.5	62.4	7.8	39
61	8	69E	5.0	B	7/7	4/3	245.3	60.5	9.5	27
62	90	75N	12.5	B	7/6	--	249.2	66.2	7.0	94
63	42	90	4.0	B	5/3	3/3	263.5	63.0	16.1	18
63a	25	90	5.0	A	4/4	--	190.4	64.9	8.4	120
64	40	90	>120	A	7/5	1/1	155.3	61.7	7.7	77
65	98	66S	>120	A	7/7	--	188.6	58.1	6.0	103
CR2		granulites		D		16/11	30.2	-53.8	13.9	12
					Paint Lake					
66-73		reworked granulites		OP		36/16	200.7	46.6	10.4	14
					Cauchon Lake					
74	12	82E	3.3	C	7/6	--	124.6	-63.9	8.9	58
75	10	60E	24.0	UN	6/0	2/0				
76	353	80W	16.0	B	7/7	--	243.3	70.0	5.4	128
77	--	--	>13.0	B	6/6	--	241.4	67.2	2.7	609
78		gneisses		UN		4/0				
79	353	67W	1.9	B	7/6	1/0	259.5	65.9	7.2	88
80	351	90	>15.0	C	6/6	2/0	92.8	-65.2	4.8	196
81	27	65W	27.1	C	9/6	--	77.2	-63.0	13.0	28
82	352	75W	4.4	C	7/3	9/4	82.0	-71.4	9.8	39
83	50	79W	>6.4	B	4/2	--	260.0	64.9*		

Table 1. (continued)

Site	*S*, deg	*D*, deg	*W*, m	RM	N_d/n_d	N_c/n_c	*D*, deg	*I*, deg	α_{95}	κ
				Cauchon Lake (continued)						
84	14	85E	24.3	C ?	6/1	1/0	109.4	-24.0*		
85	27	90	>21.0	C	7/6	2/0	100.3	-63.0	7.8	75
86	345	75E	>43.0	C	6/6	5/5	85.4	-66.1	5.3	76
87	10	80W	24.0	C	8/5	12/0	44.8	-60.7	21.7	13
88	12	--	63.5	B	6/6	1/1	237.3	61.7	8.0	58
89	22	90	2.9	B	7/7	--	226.6	63.4	5.8	109
90	80	85N	15.1	C	7/6	--	116.9	-59.6	9.3	53
91	65	74N	36.0	C	7/7	--	120.0	-59.0	5.5	122
92	349	64W	5.3	B	5/4	--	231.3	64.0	17.2	29

S, *D*, and *W*, dike strike, dip, and width; RM, remanent magnetization designation (A, B, C, or D); ER, stable magnetization components but erratic in direction; UN, unstable; OP, Trans-Hudson overprint; N_d/n_d, number of dike samples measured/ yielding stable end points at a site; N_c/n_c, number of host rock samples measured/ yielding stable end points at a site; *D* and *I*, site declination and inclination of the stable remanent magnetization; α_{95}, radius of the 95% confidence circle about the site mean direction; κ, precision of the site mean direction. CR1 and CR2, granulite samples collected in areas remote from dikes at Landing and Pikwitonei Lakes.

*Rejected site (sites accepted if N≥3 and $\alpha_{95} < 25°$).

†The dike width is the cumulate value of more than one dike.

Laboratory Methods

At least two cylindrical specimens were cut from each core or drilled from each block sample. Usually, one specimen from each sample was subjected to stepwise alternating field (AF) demagnetization up to 100 mT using a Schonstedt GSD-1 instrument, and the remanent magnetization was measured on a modified DIGICO spinner magnetometer. About one third of duplicate specimens were subjected to thermal demagnetization in at least 11 steps up to 625°C using a Schonstedt TSD-1 demagnetizer.

Components of magnetization were analyzed using orthogonal vector plots [*Zijderveld*, 1967], and their directions were determined using principal component analysis [*Kirschvink*, 1980] with a maximum angular deviation of <10° (mostly <5°) and number of data points ≥ 3. Magnetization directions of specimens from the same sample that disagreed by more than 30° were rejected; otherwise, data were combined to give sample mean directions. Finally, dike/site mean and overall mean directions were calculated using *Fisher's* [1953] statistics.

Analyses of Magnetization Components

Molson Dikes and Host Granulites

Figure 2 shows typical AF and thermal demagnetization results from dikes. Most samples contain a low-coercivity (≤15 mT) and a more stable high-coercivity component. The low-coercivity component has low to intermediate unblocking temperatures (20°-500°C) and often displays a direction close to that of the present Earth's field (PEF: D=5.8°/I=79.9°; e.g., Figures 2c, 2d, 2g, 2k, and 2l), suggesting that it is a recent viscous remanent magnetization (VRM). After removal of the VRM, a stable remanence is defined, giving linear trajectories of magnetization towards the origin of the orthogonal plots. Based on differences of magnetization directions, the stable remanence comprises three components: A (south declination with a steep downward inclination), B (WSW declination with a steep downward inclination), and C (ENE-ESE declination with a steep upward inclination). All three components have an intermediate to high-coercivity range (20-100 mT). Their thermal decay curves are all square-shouldered with unblocking temperatures typically between 550 and 580°C. These features suggest that relatively pure single/pseudo-single domain magnetite is the dominant remanence carrier. Duplicate specimens respond to AF and thermal demagnetizations in similar ways (e.g., Figures 2a/2b, 2e/2f, and 2i/2j), indicating that all three components are reliable.

The majority of dikes in the Pikwitonei Domain give excellent results on demagnetization. However, at Molson Lake and particularly at Cross Lake, rocks with unstable or erratic magnetizations are pervasive (see Table 1). This feature was also reported from other areas of the lower-grade terrain of the Superior Province and was attributed to magnetite produced from varying degrees of deuteric hydrothermal alteration of olivine or orthopyroxene [*Ermanovics and Fahrig*, 1975]. The reason why remanent magnetizations in the Molson dikes are best preserved in the Pikwitonei Domain is probably due to the relatively dry state of the host rocks. As a consequence, deuteric alteration of the dikes is either reduced or absent.

Host granulites always carry the same stable component as that of their adjacent dikes up to about 1-2 dike widths away from their contacts (Figures 3a-3f). Thermal decay curves of host rocks near contacts show square-shouldered patterns with narrow unblocking temperatures close to the Curie point of magnetite (Figures 3b, 3d, and 3f). These results suggest that both dikes and baked host rocks acquired their remanence at similar temperatures and times. Farther away, however, a consistent predike magnetization component D is observed regardless of what remanence is carried by the dike (Figures 3g-3i). Although C-like in direction, this component has a northeast rather than east declination

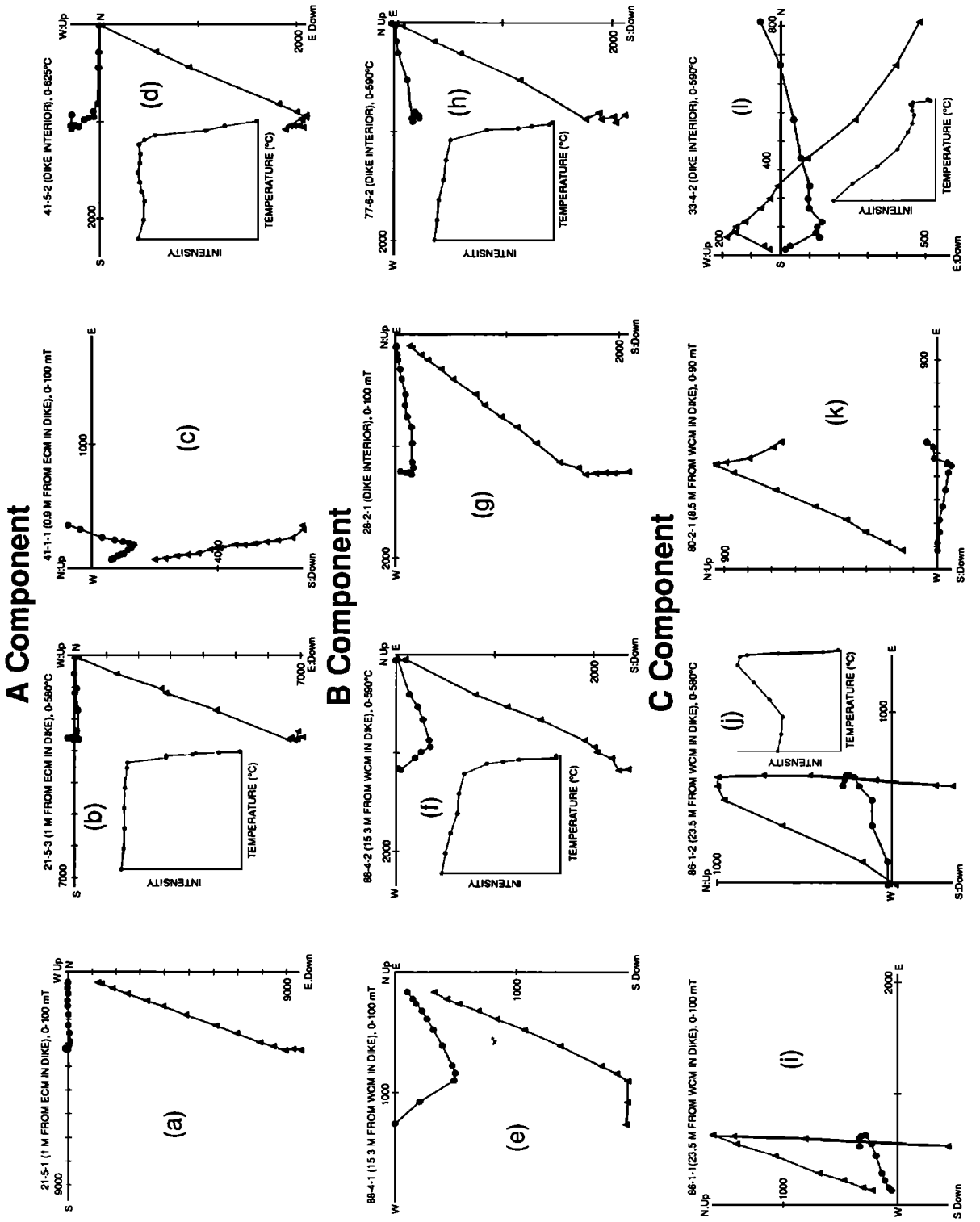


Figure 2. Orthogonal vector plots of AF and thermal demagnetization showing A, B, and C components for dikes. Solid circles (triangles) represent projections on the horizontal (vertical) plane. Units on axes are in A/m × 10^{-3}. WCM and ECM refer to the western and eastern chilled margins of dikes, respectively.

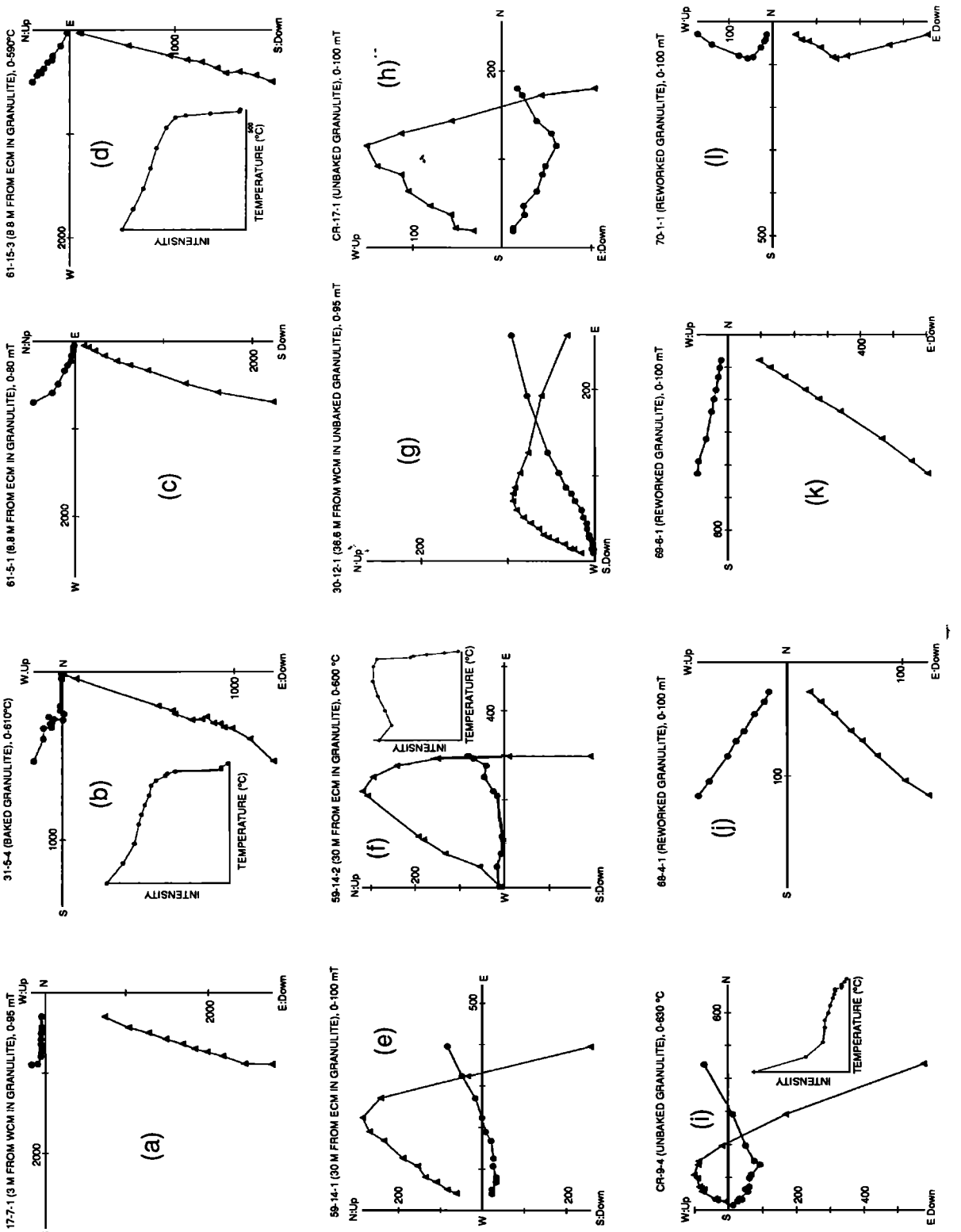


Figure 3. Orthogonal vector plots of AF and thermal demagnetization for baked (Figures 3a to 3f), unbaked (Figures 3g to 3i) and reworked Pikwitonei granulites from Paint Lake (Figures 3j to 3l). Symbols are as defined in Figure 2.

and is statistically different from C at a 95% confidence level. In addition, these unbaked host rocks exhibit a broad range of unblocking temperatures (e.g., Figure 3i).

Generally, A, B, and C do not coexist in the same dike. However, one of them may appear as an intermediate component and remain until a more stable one is isolated (Figure 4). These intermediate components are not a recent VRM because they generally have higher coercivities (15-30 mT) or unblocking temperatures (250-550°C) than that of the VRM observed in other samples. Five types of dual component behaviour upon demagnetization have been recognized: A to B (Figure 4a), A to C (Figures 4b and 4c), B to C (Figure 4d), C to D (Figure 4e), and B to A (Figure 4f). The first three are dominant in both dikes and baked host rocks.

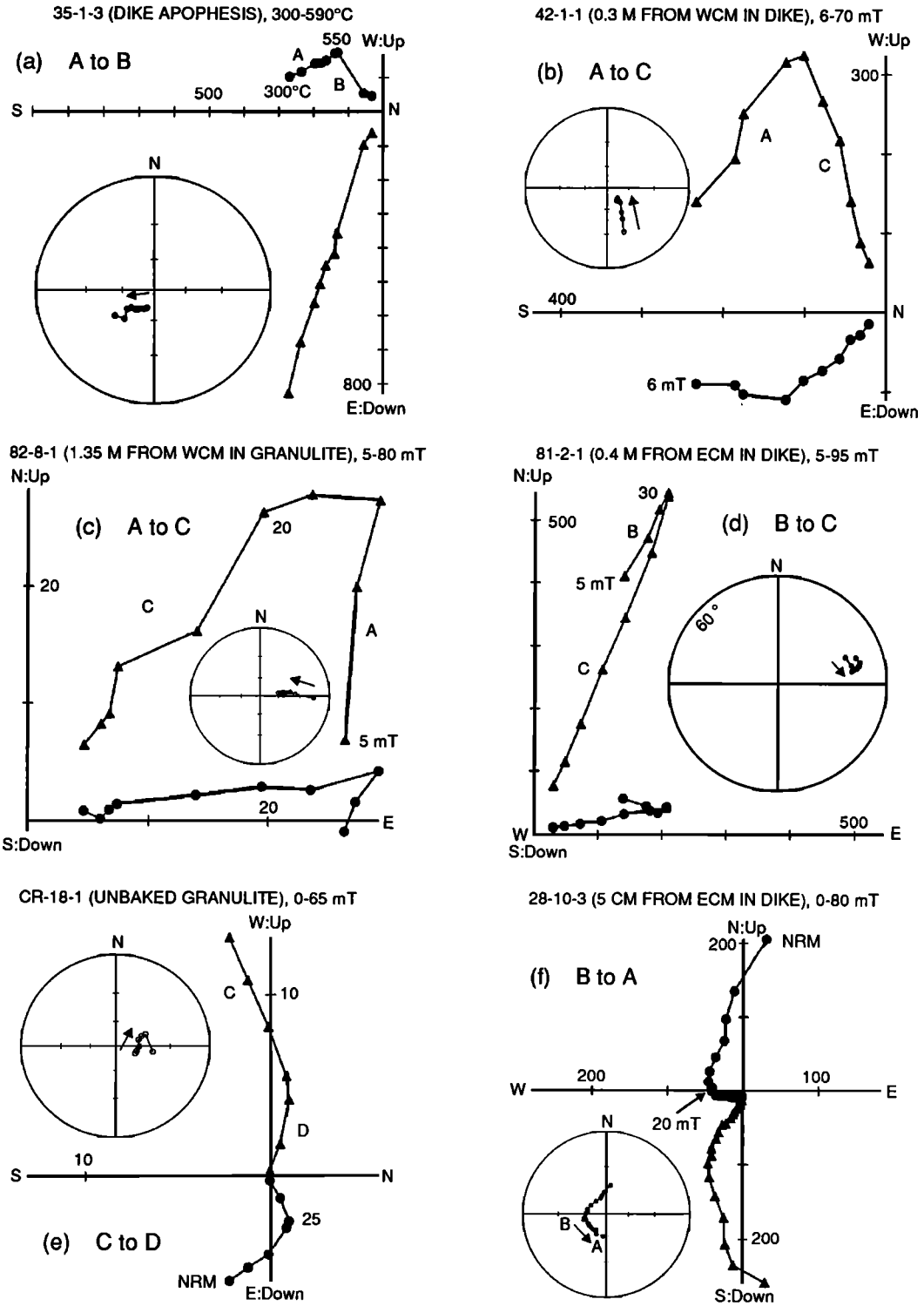


Figure 4. Orthogonal vector plots of AF and thermal demagnetization for dikes and granulites showing various combinations of overlapping magnetization components. Symbols as defined in Figure 2. Equal-angle projections of demagnetization results are also shown with solid (open) circles projected on the lower (upper) hemisphere.

This may suggest an overprinted A on B or C or an overprinted B on C. Unblocking temperature and coercivity spectra of these intermediate components show little overlap with those of the end-point components, indicative of partial thermal overprinting [*McClelland-Brown*, 1982]. C to D is only found in a few unbaked hosts and may indicate an older age for D. B to A is interpreted as a chemical overprint of A on B because A strongly overlaps B (a typical if not diagnostic feature of chemical overprinting) and samples generally show more alteration and lower intensity of NRM than those that exhibit only the B component from the same dike (see Figures 2g and 4f).

Figure 5 shows the regional distribution for A, B, and C components. Two important features are apparent. First, the same component was observed at different sites along the same dike, such as A from sites 14, 15, 17, 19, and 21 at Sipiwesk Lake and B from sites 7-10 at Cross Lake, an indication of the primary nature of these components. Second, A is mainly found in areas close to the Thompson Belt, while B and C are farther away to the east.

Reworked Granulites

Samples from amphibolitized granulite at Paint Lake are either unstable or too weak (0.01-0.1 A/m) to produce stable end-points before their NRM intensities fall below the precision of measurement. Severe destruction of magnetite and the breakdown of pyroxenes to hornblende-quartz-biotite aggregates in these rocks during Trans-Hudson metamorphism [*Russell*, 1981; *Bleeker*, 1990] are thought to be responsible for the weak and unstable magnetizations. The more mafic gneisses that have stronger magnetization intensities (0.25-1.2 A/m) exhibit a single, well-defined stable component with SSW declination and shallow to intermediate downward inclination (Figures 3j-3l). The component is isolated from about half the samples collected, and is distinct from the A, B, C, and D components. *Russell* [1981] has estimated that the temperature and pressure at Paint Lake during Trans-Hudson metamorphism were 650 ± 50° and 400-500 MPa, respectively. These conditions are high enough to destroy any previous remanence preserved in magnetite. Since the Trans-Hudson Orogen is the last major tectonic event in the region, this component is therefore most likely to be a thermal/thermochemical overprint acquired during a period between peak metamorphism at ~1820 Ma and subsequent cooling by ~1770 Ma [*Machado*, 1990; *Weber*, 1990].

Site mean and overall mean data for the Molson dikes and granulites are listed in Tables 1 and 2, and their directions are shown in Figure 6. It can be seen that components A to D are significantly different from that at Paint Lake, suggesting that they were not formed during the Trans-Hudson Orogen.

Ages of Magnetization Components

The primary nature of remanence can be established using the baked contact test, the criteria for which are those adopted by *Buchan and Halls* [1990].

The primary natures of A, B, and C are indicated by positive baked contact tests for dikes at Landing and Pikwitonei Lakes against Pikwitonei granulite host rocks, as shown by the same directions observed in both dikes and baked host rocks within about 1-2 dike widths from contacts in contrast to a stable but distinct direction D in unbaked ones (Figure 7). The square-shouldered thermal decay curves for both dikes and baked hosts in Figures 2 and 3 are in harmony with these results which suggest that all three components were formed during the original cooling of the dikes.

At the western end of Landing Lake, a dike more than 50 m wide that carries A (site 41) can be correlated with the 1883 ± 2 Ma Cuthbert Lake dike, on the basis of reprocessed federal-provincial aeromagnetic data (Figure 8). It is therefore concluded that A correlates with 1883 Ma dike intrusion.

Precise dating for B is hampered by the lack of U-Pb geochronological data. The following observations, however, suggest that B is older than A. (1) Magnetization directions of A and B are closely grouped. They show a circular Fisher distribution with no streaking (Figure 6a). A and B are therefore not caused by secular variation. (2) At eastern Pikwitonei Lake, an A dike of thickness 0.5 m is found about 2.2 m east of two B dikes with widths of 4 m and 5 m (Figure 9). A is younger than B; otherwise, the remanence of the thin dike would be reset by the intrusion of the thicker B dikes. (3) Although B dikes trend more northeastly than A dikes, the difference between their mean trends is only about 16°. This is not enough to attribute the 70° difference in the declinations of A and B to local vertical axis rotation. Consequently, two distinct geological events are responsible for the occurrence of the A and B components.

The direction of C, although almost antiparallel to B, is statistically different from B at a 95% confidence level using *McFadden and Lowes'* [1981] method. A probable explanation is that both components were acquired in normal and reversed intervals of Earth's magnetic field that might cover a relatively short period of time, probably between 2170 and 2120 Ma since poles derived from B and C lie within a group of well-dated 2170-2120 Ma poles for the Superior Province (Table 3 and Figure 10). The relative ages of B and C are unknown at present. However, since the 2200-2100 Ma apparent polar wander path for the Superior Province has a northwestward trend [*Buchan et al.*, 1994], the locations of the B and C poles would imply a slightly older age for B. Alternatively, since B coexists with C as an intermediate component in some samples (e.g., Figure 4d), C might be slightly older than B and was formed shortly after the granulites acquired D at about 2180 Ma (see the following discussion). In either case, B and C should have similar ages. In order to precisely constrain ages for B and C, samples of pegmatoidal patches from typical B and C dikes have been selected for U-Pb geochronological studies. The results will be presented in a separate paper.

D predates all three components and represents the characteristic remanence of unbaked granulites. Its age must be younger than 2640 Ma, the time of peak metamorphism in the Pikwitonei Domain, with conditions of 750°C/700 MPa at Cauchon Lake and 830°C/750-800 MPa in the center of the domain [*Mezger et al.*, 1989, 1990]. The domain then cooled slowly to ~2180 Ma when the youngest metamorphic minerals of Fe-Ti phases were formed at closure temperatures of 300-400°C. This temperature interval coincides well with the lower range of unblocking temperatures for the D component (e.g., Figure 3i), suggesting an age of ~2180 Ma for D. Furthermore, when the D pole is compared with high quality and well-dated 2220-2100 Ma poles for mafic rocks of the Superior Province (Table 3 and Figure 10), its 95% confidence oval overlaps that of the 2167 Ma pole and falls on a northwest track connecting the 2214 Ma and 2167 Ma

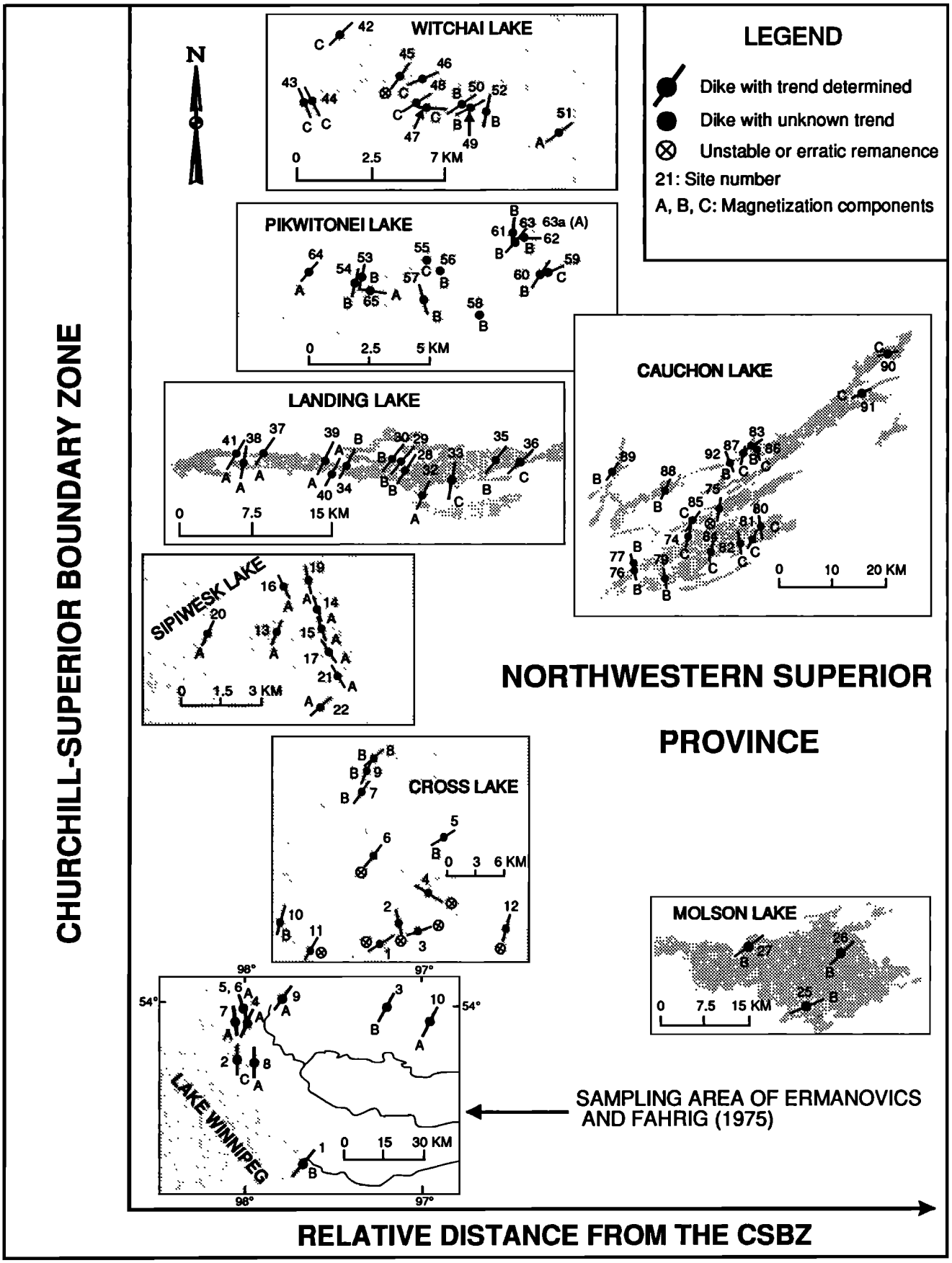


Figure 5. Schematic diagram showing spatial distribution of the A, B, and C dikes in each sampling area with respect to relative distance from the CSBZ. Sampling sites of *Ermanovics and Fahrig* [1975] are also shown with assigned magnetization classification.

Table 2. Overall Mean Paleomagnetic Data from Dikes and Pikwitonei Granulites

Location	N	D, deg	I, deg	α_{95}	κ	λ_p	ϕ_p(W)	A_{95}	K
				A Component					
Sipiwesk Lake	6	180.3	68.0	4.2	250	16.1N	98.4		
Landing Lake	6	177.4	68.2	3.9	299	16.4N	95.6		
Pikwitonei Lake	3	178.2	62.5	15.1	68	9.5N	95.7		
Witchai Lake	1	174.1	70.7	7.6	54	21.1N	93.4		
Overall Mean	16	178.5	67.2	2.7	186	15.4N	96.5	4.1	82
				B Component					
Cross Lake	4	246.3	68.4	11.8	62	29.6N	138.6		
Molson Lake	3	243.1	71.6	15.5	64	31.8N	132.5		
Landing Lake	5	257.9	60.7	9.3	68	27.1N	152.4		
Witchai Lake	3	234.1	62.9	4.8	663	20.2N	135.1		
Pikwitonei Lake	9	249.1	63.8	3.4	235	26.5N	144.1		
Cauchon Lake	6	239.4	65.7	4.7	208	24.7N	136.0		
Overall Mean	30	246.5	65.1	2.6	100	27.1N	140.8	3.8	48
				C Component					
Landing Lake	2	59.3	-58.3			16.7N	141.6		
Witchai Lake	6	114.4	-70.5	11.9	32	54.1N	160.9		
Pikwitonei Lake	2	86.5	-58.0			29.2N	160.3		
Cauchon Lake	9	95.2	-65.4	7.5	48	39.9N	157.9		
Overall Mean	19	94.2	-66.4	6.1	31	40.7N	156.3	9.7	13
				Unbaked Granulites					
Pikwitonei Domain	(30)	53.6	-64.4	6.5	13	20.9N	133.6	7.6	11
				Retrogressed Granulites					
Paint Lake	(16)	200.7	46.6	10.4	14	4.8S	115.9		

N, number of sites representing dikes (granulite samples); D/I, declination/inclination of the mean direction; α_{95}/κ, radius of the 95% confidence circle about/precision of the mean direction; λ_p/ϕ_p, latitude/longitude of the mean paleomagnetic pole; A_{95}/K, radius of the 95% confidence circle about/precision of, the mean paleomagnetic pole. Poles of overall means were calculated from virtual geomagnetic poles of individual sites.

poles. This comparison further brackets the age of the D component to a range of 2200-2170 Ma, agreeing well with the age of ~2180 Ma derived independently.

Discussion and Conclusions

Comparison of New Data and Previous Results

Our results are significantly different from those obtained by *Ermanovics and Fahrig* [1975] for Molson dikes in more southerly areas (Figure 6b). Examination of their data shows that all three components identified in this study are present (A, their sites 4 to 9 and 10; B, sites 1 and 3; C, site 2), but directions were averaged as if they all belonged to the same age of magnetization. If we look at the regional distribution of these components (Figure 5), a similar pattern to ours is found with A mainly occurring in areas close to the Thompson belt and B and C farther to the east.

Multiple Episodes and Spatial Distributions of Dike Emplacement

A major discovery of this study is the identification of three primary components A, B, and C from the traditional Molson dike swarm. Among them, A is well constrained at 1883 Ma by U-Pb zircon ages [*Heaman et al.*, 1986a], B and C are estimated at 2170-2120 Ma on the basis of their distinct paleomagnetic characteristics from the A component, the analysis of metamorphic history of the Pikwitonei Domain, and comparison with known reliable paleomagnetic poles for the Superior Province. Therefore some dikes in the swarm may have been emplaced well before the 1883 Ma dike intrusion. This is contrary to the current view of emplacement only at 1883 Ma which has been derived largely on the basis of U-Pb zircon ages from two dikes in the swarm [*Heaman et al.*, 1986a]. Since one of the dated dikes has an A component, these U-Pb zircon ages only define an episode of A dike emplacement and cannot be regarded as representative for all of the dikes in the swarm. As a consequence, the discovery of the three primary magnetic components provides the first supporting evidence for the multiple age of the swarm that *Corkery* [1985] determined from geological evidence in the Split Lake area.

The similarity in pole positions of the B and C dikes suggests that they are part of the same swarm but emplaced at times of opposite ambient field polarity. In the following discussions, we will refer to these dikes as the Cauchon swarm after Cauchon Lake where B and C dikes occur in

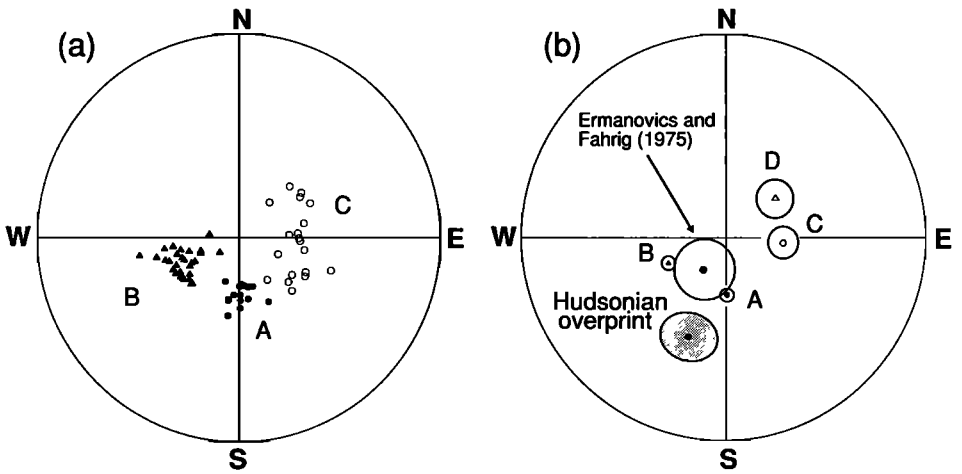


Figure 6. Equal-area projections of (a) site mean directions for dikes and baked granulites, and (b) overall mean ovals of 95% confidence for the A, B, C, and D components. The Paleoproterozoic Trans-Hudson overprint from Paint Lake rocks and the magnetization direction of *Ermanovics and Fahrig* [1975] are shown in the diagram for comparison. Solid (open) symbols represent projections onto the lower (upper) hemisphere.

relative abundance. Since the age of the younger A dikes is constrained at ~1883 Ma on the basis of U-Pb dating from the Cuthbert Lake dike [*Heaman et al.*, 1986a], we will refer to these dikes as the Cuthbert swarm.

The dikes of the Cuthbert swarm are (1) contemporaneous with or slightly earlier than the Trans-Hudson Orogen; (2) more parallel to the Thompson Belt than either B or C dikes; and (3) predominant in areas close to the Thompson Belt. These features indicate that the swarm is most likely related to the development of the Trans-Hudson Orogen. Furthermore, this swarm may be correlated to some mafic igneous rocks within the CSBZ on the basis of similar U-Pb zircon ages or structural relationships. For example, the east trending Fox River sill in the Fox River Belt has yielded a U-Pb zircon age of 1883 ± 1.5 Ma [*Heaman et al.*, 1986a], clearly indicating that it is comagmatic. In the Split Lake area, the younger dikes are east-trending [*Corkery*, 1985], parallel to the structure of the Split Lake Block. This would

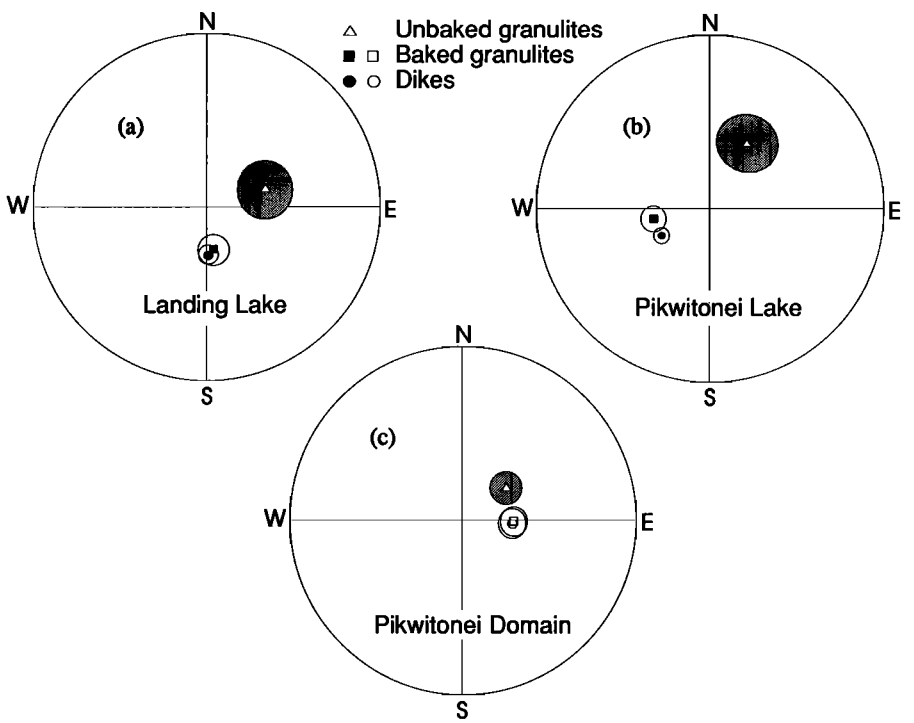


Figure 7. Positive baked contact tests for the A, B, and C components. Ovals of 95% confidence are shown for each direction. Other symbols are as defined in Figure 6. Baked granulites comprise samples within about 1-2 dike widths from contacts and unbaked ones represent samples at least 2 dike widths away.

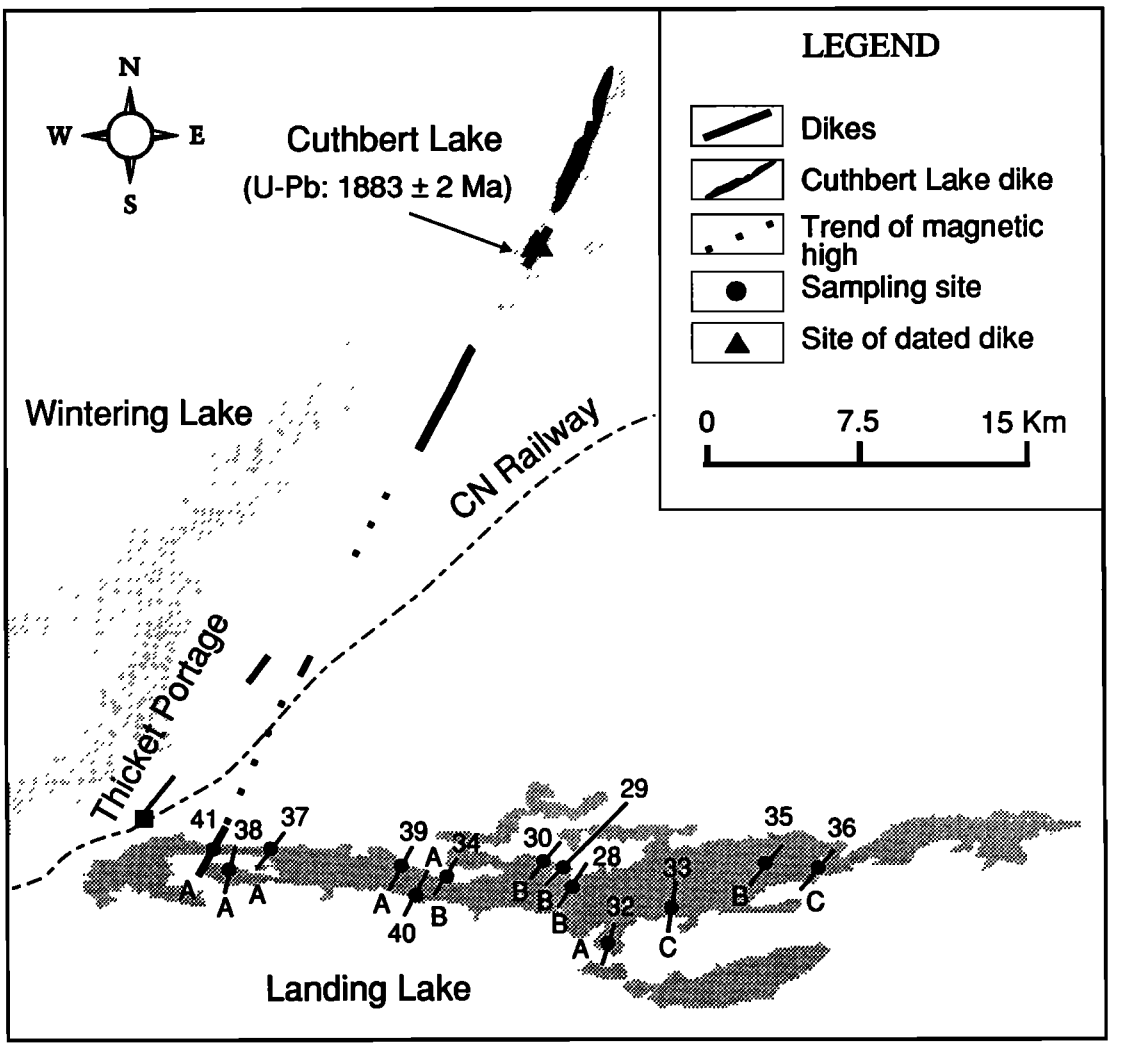


Figure 8. Map showing sampling sites in Landing Lake and the trend of magnetic anomalies that can be traced from site 41 to the 1883 Ma Cuthbert Lake dike. A, B, and C represent magnetization components.

be expected if the younger dikes are correlated with the Cuthbert swarm. The earlier dikes in the Split Lake area, however, could be extensions of the Cauchon swarm because they show a similar NE-trend, a characteristic of pre-1883 Ma dikes. Mafic dikes in the Thompson Belt can also be correlated to the Cuthbert swarm since they exhibit similar NNE-NNW trends and appear to postdate the earliest Trans-Hudson fold structures [*Bleeker*, 1990]. Consequently, the Cuthbert swarm is mainly constrained within the CSBZ and along areas close to the Thompson Belt in the northwestern margin of the Superior Province.

The age, orientation, and spatial distribution of the 1883 Ma Cuthbert swarm all suggest that its emplacement is closely related to the Trans-Hudson Orogen. The model of a foredeep setting [*Hoffman*, 1988] seems unlikely, because there is no stratigraphic evidence of advancing thrust sheets from the Reindeer Zone in the metasedimentary rocks of the CSBZ [*Weber*, 1990], and recent seismic profiles show a prominent east dipping reflection boundary in the southeastern Trans-Hudson Orogen [*Nelson et al.*, 1993; *Lewry et al.*, 1993; *White et al.*, 1994]. We prefer a back arc spreading setting for the emplacement of the Cuthbert swarm [*Weber*, 1990] because (1) it has rift-type chemical affinities [*Baragar and Scoates*, 1981; *Scoates*, 1981], (2) its age is either slightly younger or equal to 1910-1878 Ma arc volcanic rocks caused by subduction of Paleoproterozoic oceanic crust in the Reindeer Zone [*Baldwin et al.*, 1987; *Syme*, 1990], and (3) new trace element data from coeval mafic magmatic rocks of the CSBZ indicate a depleted mantle source for those rocks [*Halden*, 1988].

Despite their paleomagnetic differences, the A, B, and C dikes cannot be differentiated by major element geochemical data [*Scoates and Macek*, 1978; *Ermanovics and Fahrig*, 1975]. This geochemical uniformity is confirmed by major element analyses (X-Ray Assays, Don Mills, Ontario) on 22 near-margin samples of dikes chosen equally from A, B, and C populations (Y. Zhai and H.C. Halls, unpublished data, 1992).

Uplift and Initial Rifting of Northwestern Margin of the Superior Province

Several geologists [e.g., *Weber*, 1990; *Bleeker*, 1990] have proposed that the Pikwitonei granulites were brought to shallow depths before the 1883 Ma dike intrusion because

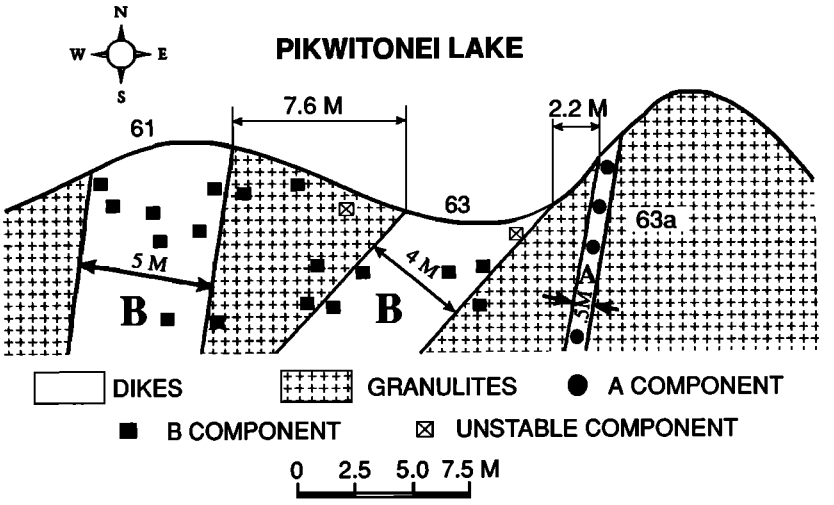


Figure 9. Sample locations in two B dikes (sites 61 and 63) and an adjacent A dike (site 63a) showing that dikes 61 and 63 are older than 63a.

(1) glassy contacts between dikes and hosts were observed in both the Pikwitonei Domain and the lower-grade terrain of the Superior Province and (2) the Ospwagan Group of metavolcanic and metasedimentary rocks in the Thompson Belt was deposited on reworked Pikwitonei granulites and cut by probable Cuthbert dikes. Our study confirms this proposal and indicates that these glassy contacts can be observed not only in the 1883 Ma A dikes but also among earlier B and C dikes. This suggests that the major part of the uplift of the northwestern margin of the Superior Province had occurred before the emplacement of the Cauchon swarm and, as a result, may be responsible for the occurrence of the D component in the granulites.

This uplift is approximately coeval in time with the prevalent granulite facies metamorphism that occurred between 2300-2000 Ma in the Western Granulite domain of Rae Province, which may represent eastward overthrusting of a Tibetan-type uplift zone [*Bickford et al.*, 1994] or, alternatively, the first stage of tectonic developments in the model of initial rifting [*Green et al.*, 1985] that might be related to the opening of *Stauffer's* [1984] Manikewan ocean. If our age estimate for B and C dikes is correct, their emplacement would be about 200 Ma earlier than the oldest 1910 Ma arc-related volcanism found in the Reindeer Zone [*Baldwin et al.*, 1987]. This would imply that the emplacement of the Cauchon swarm is not related to the Trans-Hudson Orogen, but rather to an earlier rifting event.

Young [1973, 1988] proposed that Proterozoic glacigenic rocks, such as the Huronian Supergroup of the Superior Province, Hurwitz Group of the Hearne Province, and Snowy Pass Supergroup of the Wyoming Province, can be stratigraphically correlated. He further suggested that these rocks were formed in marine basins in an extensional tectonic regime and that they developed in an advanced stage of the continental rifting process corresponding to the opening of the Manikewan Ocean between the Hearne and Superior Provinces. The Cauchon swarm, therefore, can be correlated with rift-related mafic rocks in the lower Hurwitz Group of the Hearne Province which have a U-Pb baddeleyite-zircon age of 2094 +26/-17 Ma [*Patterson and Heaman*, 1991].

In conclusion, the following Paleoproterozoic history for the northwestern Superior Province is proposed. (1) The northwestern Superior Province was first uplifted about 2200-2170 Ma. The D component may have formed at this time,

Table 3. 2100-2220 Ma Paleomagnetic Poles for the Superior Province

Units	Age, Ma	λ_p	ϕ_p (W)	α_{95}	Test	Reference
Nipissing N1	U2217±4	16.5N	85.2	7.8	c	*Buchan et al.* [1994]
Senneterre dikes	U2214±13	15.3S	75.7	5.5	c	*Buchan et al.* [1994]
Biscotasing dikes	U2167±2	27.8N	136.7	8.1	c	*Buchan et al.* [1994]
Fort Frances dikes	R2120±67	43.0N	176.0	5.1	s	*Halls* [1986]
Marathon dikes (N)	U2113+17.2/–6.5	42.8N	164.4	6.4	–	*Halls et al.* [1994]
Marathon dikes (R)	2110-2120	51.2S	5.3	6.8	c	*Halls et al.* [1994]
Molson dikes (B)	2120-2170	27.1N	140.8	3.8	c	this study
Molson dikes (C)	2120-2170	40.7N	156.3	9.7	c	this study

U, U-Pb zircon age; R, Rb-Sr age; λ_p/ϕ_p, latitude/longitude of paleomagnetic pole; α_{95}, radius of the 95% confidence circle about mean magnetization direction from which the paleomagnetic pole is derived; c, positive baked contact test; s, secular variation from dike to dike.

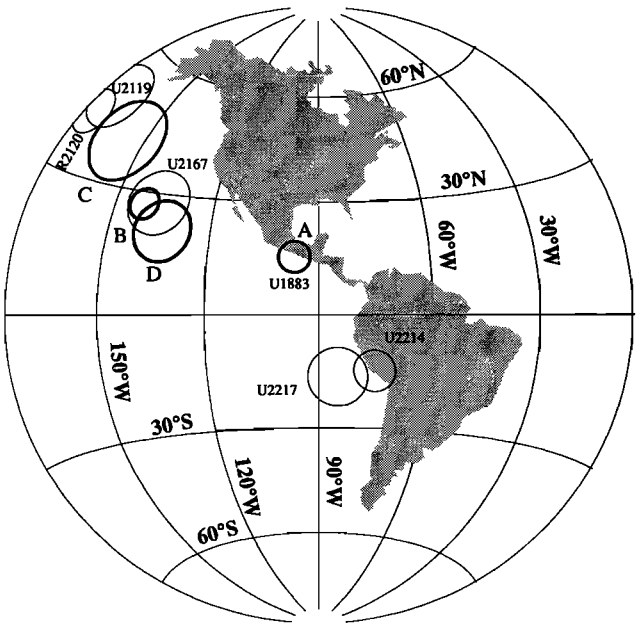


Figure 10. Orthographic azimuthal projection showing paleopole positions for the A, B, C, and D components compared with those for high quality 2200-2120 Ma paleopoles for the Superior Province [see *Buchan et al.*, 1994].

although isobaric cooling cannot be discounted. (2) The subsequent rifting under an extensional tectonic setting along the northwestern Superior and southeastern Hearne Provinces opened the Manikewan Ocean. During this process, mafic magma from the upper mantle intruded along the newly-rifted continental margins, forming gabbro sills in the lower Hurwitz Group and the Cauchon swarm in the northwestern Superior Province. The emplacement of the swarm occurred in normal and reversed geomagnetic fields within a period of 2170 to 2120 Ma when the B and C components were formed. (3) During the 1910-1870 Ma interval, eastward subduction of the Manikewan Ocean beneath the Superior Craton caused large scale felsic arc-volcanism in the Reindeer Zone of the Trans-Hudson Orogen, and eventually led to the formation of a back arc basin [*Weber*, 1990] where the Cuthbert swarm was intruded at about 1883 Ma.

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